

Fangfei Li

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Authorized to work in the U.S. (no sponsorship required now or in the future)

SUMMARY & SKILLS

• **Ph.D. in Computational Biology, Applied Mathematics & Statistics** | 2 yrs postdoc + 2 yrs biotech startup + 3 yrs wet-lab (synthetic barcoding-based *in vivo* yeast lineage tracing, molecular biology) • **Skills:** data analysis & visualization, deep learning, bioinformatics, NGS, Bayesian optimization, Python (NumPy, Matplotlib, tkinter, Flet), Git, SQL, API, MATLAB, Computational modeling & algorithm development, • Strong analytical thinker; detail-oriented, organized, and goal-driven.

INDUSTRIAL & ACADEMIC EXPERIENCE

AI Training with Certificates ————— 9/2025–current Online

- Deep Learning Specialization – DeepLearning.AI (Coursera): [Certificate](#), [Learning Notes](#)
- Artificial Intelligence Professional Program – Stanford:
 - Deep Generative Models (XCS236): [Certificate](#)
 - Deep Learning for Computer Vision (XCS231N): [Certificate](#)
 - Natural Language Processing with Deep Learning (XCS224N): [Certificate](#)

Computational Biology Staff Scientist ————— 11/2023–8/2025

BacStitch DNA, South San Francisco, CA

- Developed an automated end-to-end DNA assembly workflow (Python, web/GUI) that scores sequences, designs optimal overlapping blocks, checks synthesizability via supplier APIs, and generates automated high-throughput experimental plans.
- Built a Python tool for optimal DNA block design that segments sequences into overlapping blocks with minimal structural features in overlaps, under customizable constraints on block count, block/overlap length, and specific regions forced as blocks/overlaps.
- Created Python visualization tools to (i) plot BLAST results and structural scores ([Example](#)), (ii) generate *.ab1* files encoding structural scores as ATGC traces at single-nucleotide resolution, and (iii) produce publication-quality figures for publications and internal use.

Postdoctoral Research Fellow ————— 4/2021–10/2023

Sherlock Lab at Department of Genetics, Stanford University, CA

- Developed a bioinformatics pipeline to process raw NGS reads, perform quality control and barcode extraction, and generate barcode count tables for downstream analysis.
- Improved a fitness estimation algorithm and developed a Python pipeline ([FitSeq2](#)), an enhanced version of [Fit-Seq](#).
- Developed a Bayesian optimization algorithm and a Python pipeline ([FitMut2](#)) to identify genome-wide adaptive lineages from fitness phenotypes (without WGS) in experimental evolution and estimate mutation fitness effects and occurrence times.

Graduate Researcher ————— 8/2014–12/2020

Laufer Center for Physical and Quantitative Biology, Stony Brook University, NY

- Developed a fitness estimation tool ([Fit-Seq](#)) that can estimate unbiased fitness for complex cell pools, using a maximum likelihood algorithm.

- Designed and conducted *in vivo* bar-seq experimental yeast evolution (yeast genetics, cloning, barcoded library construction, lineage-traced evolution, and NGS prep). Performed computational analysis (NGS processing, mathematical modeling for inferring evolutionary parameters from lineage dynamics, and evolutionary simulations for validation) to explore the epistasis of adaptive mutations in yeast evolution.

EDUCATION **Ph.D., Computational Biology, Applied Mathematics and Statistics**——8/2013–12/2020
 Stony Brook University, Stony Brook, NY
M.S., Systems Theory——9/2010–7/2013
 University of Chinese Academy of Sciences, Beijing, China
B.S., Mathematics and Applied Mathematics——9/2006–7/2010
 Beijing Normal University, Beijing, China

PRESENTATIONS **Fitness Estimation of Pooled Amplicon Sequencing Studies (Plenary Talk)**——8/2018
 Yeast Genetics Meeting, Stanford CA

PUBLICATIONS [1] T Matsui, P-H Hung, H Mei, X Liu, **F Li**, J Collins, W Li, D Miller, N Wilson, E Toro, G J Taghon, G Sherlock, S Levy. [High-throughput DNA engineering by mating bacteria](#). Preprint (2024).
 [2] **F Li**, A Mahadevan, G Sherlock. [An improved algorithm for inferring mutational parameters from Bar-seq evolution experiments](#). *BMC Genomics* 24, 246 (2023).
 [3] **F Li**^{*}, J Tarkington^{*}, G Sherlock. [Fit-Seq-2.0: an improved software for high throughput fitness measurements using pooled competition assays](#). *Journal of Molecular Evolution* 91, 334–344 (2023).
 [4] G AVECILLA, J N Chuong, **F Li**, G Sherlock, D Gresham, Y Ram. [Neural networks enable efficient and accurate simulation-based inference of evolutionary parameters from adaptation dynamics](#). *PLoS Biology* 20, e3001633 (2022).
 [5] Z Liu, D Miller, **F Li**, X Liu, S Levy. [A large accessory protein interactome is rewired across environments](#). *eLife* 9, e62365 (2020).
 [6] X Liu, Z Liu, A K Dziulko, **F Li**, D Miller, R D Morabito, D Francois, S Levy. [iSeq 2.0: a modular and interchangeable toolkit for interaction screening in yeast](#). *Cell Systems* 8, 338–344 (2019).
 [7] **F Li**, M L Salit, S Levy. [Unbiased fitness estimation of pooled barcode or amplicon sequencing studies](#). *Cell Systems* 7, 521–525 (2018).
 [8] I Frumkin, D Schirman, A Rotman, **F Li**, L Zahavi, E Mordret, O Asraf, S Wu, S Levy, Y Pilpel. [Gene architectures that minimize cost of gene expression](#). *Molecular Cell* 65, 142–153 (2017).
 [9] Y Chen, W Xiong, **F Li**. [Convergence of infinite products of stochastic matrices: a graphical decomposition criterion](#). *IEEE Transactions on Automatic Control* 61, 3599–3605 (2016).
 [10] Y Chen, **F Li**, B Hou, S Tan, H Zhu. [Convergence analysis of discrete-time consensus algorithm with both self and transmission delays](#). *Journal of the Franklin Institute* 353, 2467–2481 (2016).
 [11] **F Li**, Y Chen, J Lü, D Hill. [Cluster consensus of Boolean multi-agent systems](#). 2013 9th Asian Control Conference (ASCC) (2013).
 [12] Y Chen, W Yu, **F Li**, S Feng. [Synchronization of complex networks with impulsive control and disconnected topology](#). *IEEE Transactions on Circuits and Systems II: Express Briefs* 60, 292–296 (2013).